IDMA (matrices)

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matrices

arrays of number

pictures

linear transformations

50×50 picture

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109	110	111	110	110	110	110	110	110	110	109	110	109	108	108	108	99	72	62	60	62	63	62	64	76	92	98	101	103	104	104	104	105	104	104	104	103	104	104	103	102	101	102	101	100	100	99	99	98	96
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]

example

a matrix A is a rectangular array of numbers

if n rows and k columns (i is row and j is column)

$$A_{i,j} \in \mathbb{R} \quad i \in [n], j \in [k]$$

for example

$$A = \begin{bmatrix} 1 & 0 & -1 & 6 \\ 3 & 4 & -1 & 9 \\ 0 & 0 & 0 & \sqrt{2} \end{bmatrix}$$

with $A_{1,1} = 1$ and $A_{3,1} = 0$

rows and columns

$n \times k$ matrix A is a rectangular array of numbers

the rows are vectors in $\mathbb{R}^k$

the columns are vectors in $\mathbb{R}^n$

square matrices

$n \times n$ matrix A is a square matrix

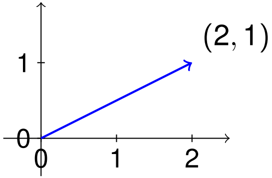
examples

zero matrix
identity matrix I_n
scalar matrix cI_n
diagonal matrix

2×1 **matrix**

a 2×1 matrix (a.k.a. column vector)

$$v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$



operations

transposition

addition

multiplication

transposition

the transpose A^T of A is

$$A^T_{i,j} = A_{j,i}$$

for example

$$\begin{bmatrix} 1 & 0 \\ 3 & 4 \\ 0 & 0 \end{bmatrix}^T = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 4 & 0 \end{bmatrix}$$

transposition: running time

the running time of computing A^T for an $n \times k$ matrix is $O(kn)$

here and below operations on numbers take time $O(1)$

transposition: properties

transposition is an involution

$$(A^T)^T = A$$

addition

if A, B are both $n \times k$ (same dimensions) then

$$(A+B)_{i,j} = A_{i,j} + B_{i,j}$$

in words, addition is pointwise

for example

$$\begin{bmatrix} 1 & 0 \\ 3 & 4 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} -1 & 1 \\ 3 & 0 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 6 & 4 \\ 2 & 1 \end{bmatrix}$$

addition: running time

the running time of computing $A + B$ for two $n \times k$ matrices A, B is $O(kn)$

for $i = 1, \ldots n$
  for $j = 1, \ldots, k$
    set $C_{i,j} = A_{i,j} + B_{i,j}$

remark running time is sharp because that is size of output

addition: properties

for $A + B$ to be defined, need to be of same dimensions

for three $n \times k$ matrices A, B, C

$$A + B = B + A$$
$$(A + B) + C = A + (B + C)$$
$$A + 0 = A$$

where 0 is the $n \times k$ zero matrix

scalar multiplication

for $c \in \mathbb{R}$ and a matrix A the matrix cA is

$$(cA)_{i,j} = cA_{i,j}$$

in words, multiply all entries by c

addition + scalars

addition and scalar multiplication make matrices into a vector space

if $n \times k$ matrices then the dimension is nk

multiplication

if A is $1 \times k$ and B is $k \times 1$ then AB is 1×1

$$AB = \sum_{\ell=1}^{k} A_{1,\ell} B_{\ell,1}$$

this is called inner product or dot product

for example

$$\begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \\ 7 \\ 8 \end{bmatrix} = 1 \cdot 5 + 2 \cdot 6 + 3 \cdot 7 + 4 \cdot 8 = 70$$

multiplication

if A is $n \times k$ and B is $k \times 1$ then AB is $n \times 1$

$$(AB)_{i,1} = \sum_{\ell=1}^{k} A_{i,\ell} B_{\ell,1}$$

n inner products

for example

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3 \\ 4 \cdot 1 + 5 \cdot 2 + 6 \cdot 3 \\ 7 \cdot 1 + 8 \cdot 2 + 9 \cdot 3 \\ 10 \cdot 1 + 11 \cdot 2 + 12 \cdot 3 \end{bmatrix} = \begin{bmatrix} 14 \\ 32 \\ 50 \\ 68 \end{bmatrix}$$

multiplication

if A is $n \times k$ and B is $k \times m$ then AB is $n \times m$

$$(AB)_{i,j} = \sum_{\ell=1}^{k} A_{i,\ell} B_{\ell,k}$$

k matrix times vector products

for example

$$\begin{bmatrix} 1 & 0 \\ 3 & 4 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 5 & 7 \\ 0 & 0 \end{bmatrix}$$

matrix times a vector: plane

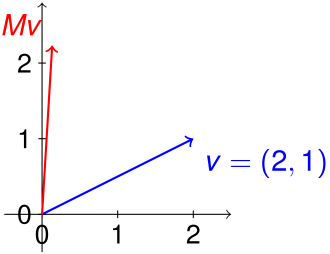
the 2×2 matrix

$$M = \begin{bmatrix} \cos\left(\frac{\pi}{3}\right) & -\sin\left(\frac{\pi}{3}\right) \\ \sin\left(\frac{\pi}{3}\right) & \cos\left(\frac{\pi}{3}\right) \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$$

is rotation in $\frac{\pi}{3} = 60$ degree in the plane

matrix times a vector: plane

$$v = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \qquad Mv = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 - \frac{\sqrt{3}}{2} \\ \sqrt{3} + \frac{1}{2} \end{bmatrix}$$



matrix times a matrix

if A is $n \times k$ and B is $k \times m$ then the i, j position of AB is the inner product between the i-row of A and the j-column of B

multiplication: running time

if A, B are $n \times n$ then

$$(AB)_{i,j} = \sum_{\ell=1}^{n} A_{i,\ell} B_{\ell,k}$$

obvious running time is

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obvious running time is $O(n^3)$

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is this sharp?

multiplication: running time

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obvious running time is $O(n^3)$

is this sharp? no!

Strassen (1969): can multiply in time $O(n^{2.81})$

... (2024) can multiply in time $O(n^{2.38})$

Strassen's idea

can multiply 2×2 matrices using 7 multiplications (instead of 8)

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running time is

$$T(n) = 7 \cdot T\left(\frac{n}{2}\right) + O(n^2)$$

$O(n^2)$ additions

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running time is

$$T(n) = 7 \cdot T\left(\frac{n}{2}\right) + O(n^2)$$

$O(n^2)$ additions

which leads to

$$T(n) = O\left(n^{\log_2 7}\right) \approx O\left(n^{2.81}\right)$$

multiplication: properties

the product AB is defined only when number of columns in A is number of rows in B

if A, B, C are such that AB, BC are defined then

$$(AB)C = A(BC)$$

commutativity does not hold: there are square A, B such that

$$AB \neq BA$$

for every $n \times k$ matrix A

$$I_k A = A I_n = A$$

operation: pictures

if A, B are square pictures what are

$$A^T, A + B, AB?$$

boolean matrices

a matrix A is boolean if for every i, j

$$A_{i,j} \in \{0, 1\}$$

operations: and or

if A, B are $n \times k$ boolean matrices then $A \vee B, A \wedge B$ are

$$(A \vee B)_{i,j} = A_{i,j} \vee B_{i,j}$$
$$(A \wedge B)_{i,j} = A_{i,j} \wedge B_{i,j}$$

operation: product

if A, B are $n \times n$ boolean matrices then $A \odot B$ is defined by

$$(A \odot B)_{i,j} = \bigvee_{\ell=1}^{n} (A_{i,\ell} \wedge B_{\ell,j})$$

compare to $\sum_{\ell} A_{i,\ell} B_{\ell,j}$

a graph

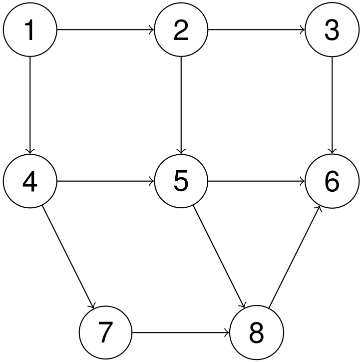
a combinatorial object that is a model for many things

social network

cities and roads

market in economy

example

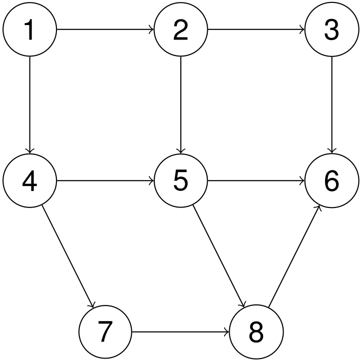


definition

a graph G is $G = (V, E)$

V is a set (vertices)

E is a set of pairs (edges)



remarks

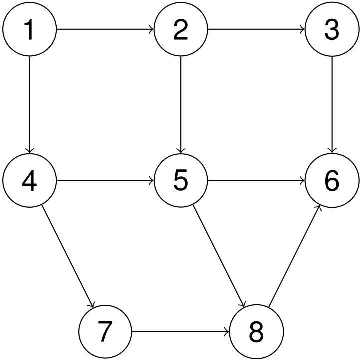
could be directed (u, v) or undirected $\{u, v\}$

distances

graphs have paths

paths have lengths

notion of “distance”

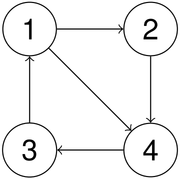


adjacency matrix

a graph $G = (V, E)$ has adjacency matrix A_G defined by

$$(A_G)_{u,v} = \begin{cases} 1 & (u, v) \in E \\ 0 & (u, v) \notin E \end{cases}$$

adjacency matrix: example



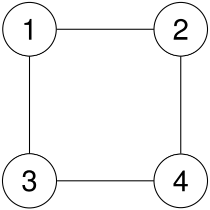
$$A_G = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

symmetry

claim

G is undirected iff A_G is symmetric ($A_G^T = A_G$)

symmetry: example



$$A_G = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}.$$

powers: definition

if A is $n \times n$ matrix then

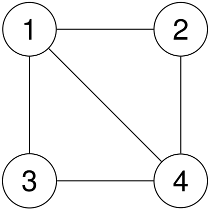
$$A^1 = A$$

and inductively define

$$A^{d+1} = A^d \cdot A$$

remark $A^0 = I_n$

powers: example



$$A_G = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \qquad A_G^2 = \begin{bmatrix} 3 & 1 & 1 & 2 \\ 1 & 2 & 2 & 1 \\ 1 & 2 & 2 & 1 \\ 2 & 1 & 1 & 3 \end{bmatrix}$$

powers

claim if G is a graph and $A = A_G$ then

$A_{u,v} = 1$ iff there is a $u \to v$ path of length one

powers

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powers

claim if G is a graph and $A = A_G$ then

$A_{u,v} = 1$ iff there is a $u \to v$ path of length one

$(A^2)_{u,v} = \sum_w A_{u,w}A_{w,v}$ is number of $u \to v$ paths of length two

$(A^d)_{u,v}$ is number of $u \to v$ paths of length d

exercise prove by induction

boolean powers

exercise if G is a graph then $A = A_G$ is boolean

what is the meaning of

$$A^{\odot d}?$$

where

$$A^{\odot 1} = A$$

and

$$A^{\odot (d+1)} = A^{\odot d} \odot A$$

summary

matrices: numbers, boolean

operations: point-wise and products

running times

graphs and their matrices (there is a lot of info here)